

From Pricing to Performance

Analyzing the Drivers of
Airline Operations and
Passenger Satisfaction

Presented By : Yellow Group 9



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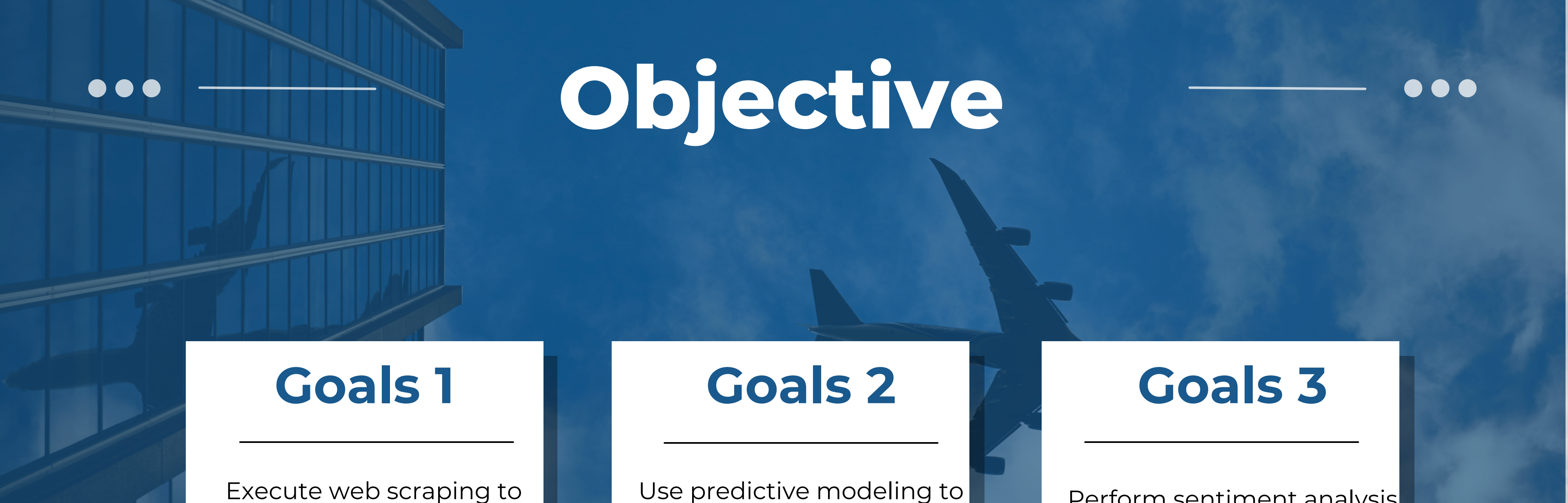


Problem Statement

- Airlines face challenges in understanding customer feedback effectively and addressing urgent customer concerns in real time.
- Operational disruptions such as flight delays and weather issues often escalate customer dissatisfaction.
- There is a lack of systems that integrate customer feedback with operational data to provide actionable insights for improving customer satisfaction.

Introduction

- Customer reviews provide valuable insights into the quality of airline services, highlighting both strengths and areas needing improvement.
- Current review systems often fail to flag and escalate urgent issues, such as prolonged delays or unaddressed complaints.
- This project proposes a real-time review analysis system that combines customer feedback and operational data to help airlines proactively address customer concerns.



Objective

Goals 1

Execute web scraping to collect text reviews and applying an LDA model to extract the key topics

Goals 2

Use predictive modeling to evaluate the relationship between external factors (e.g., weather, time of day, airport traffic, and aircraft age) and flight delays

Goals 3

Perform sentiment analysis on online airline reviews collected via web scraping to uncover public perceptions of airline services

Project Methodology

Data Collection

Web scraping to gather customer reviews from online platforms.

Data Processing

Clean and preprocess review text to identify themes and keywords.

Analysis & Visualization

Analyze patterns in operational data, such as delay trends by time of day or weather conditions.

Insights

Deliver actionable recommendations for airlines based on data analysis.

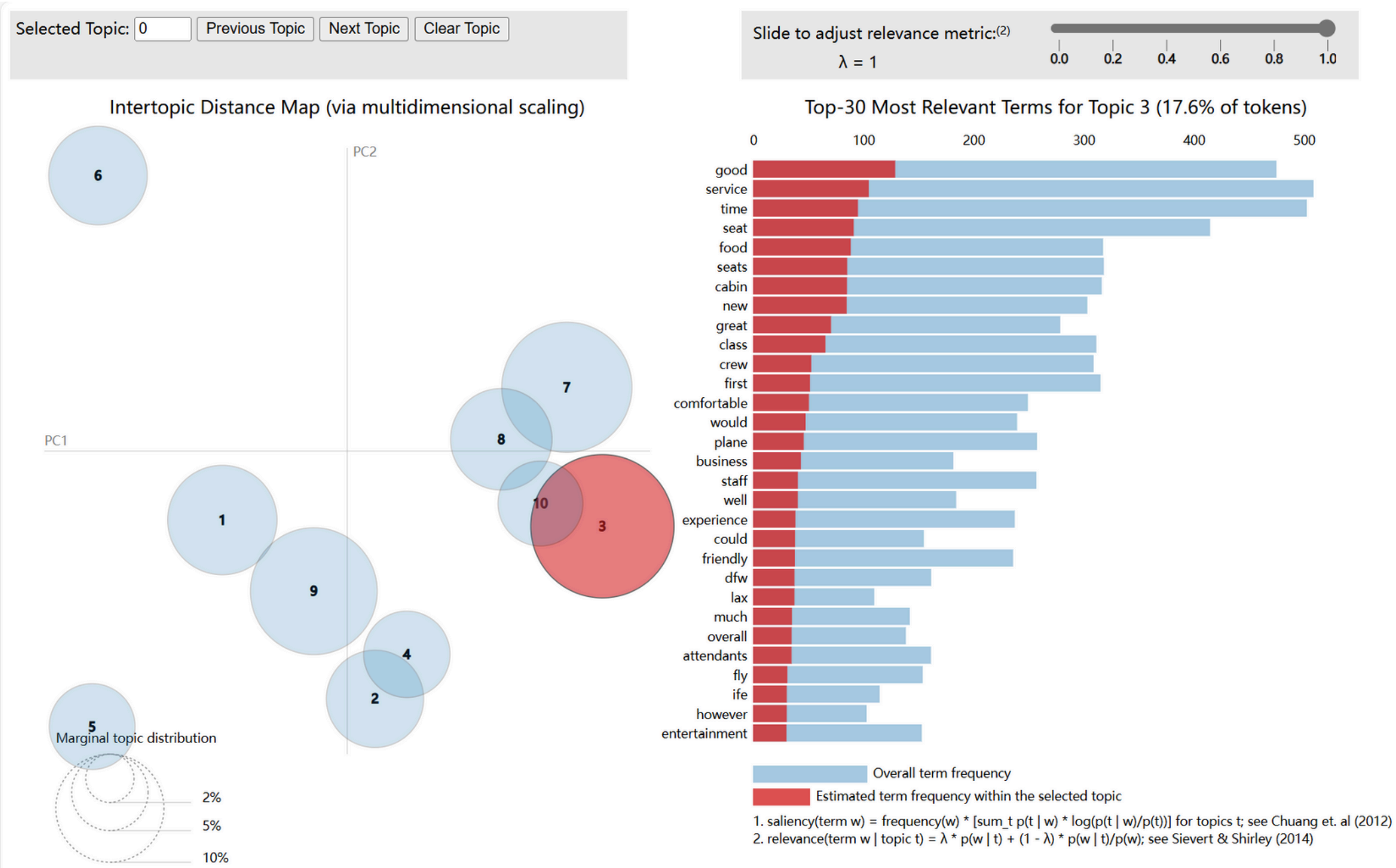
Data at a Glance

Overview of Collected Data

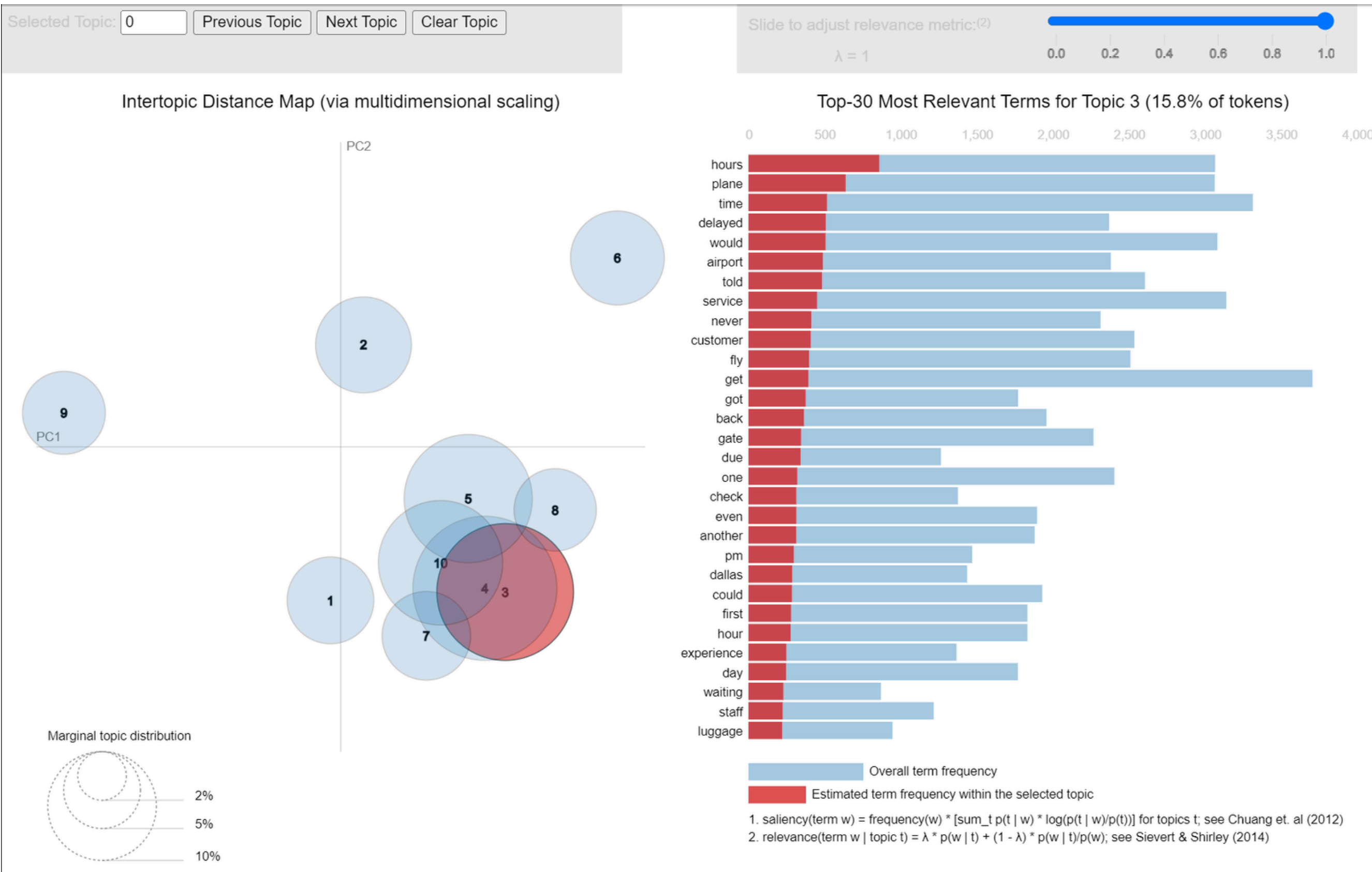
- Data is scraped from Skytrax, including text reviews, review dates, recommendations, etc.
- Library used:
 - Data scraping: BeautifulSoup
 - Text analysis: Gensim, nltk, pyLDAvis



Analysis - Positive Reveiw Topics

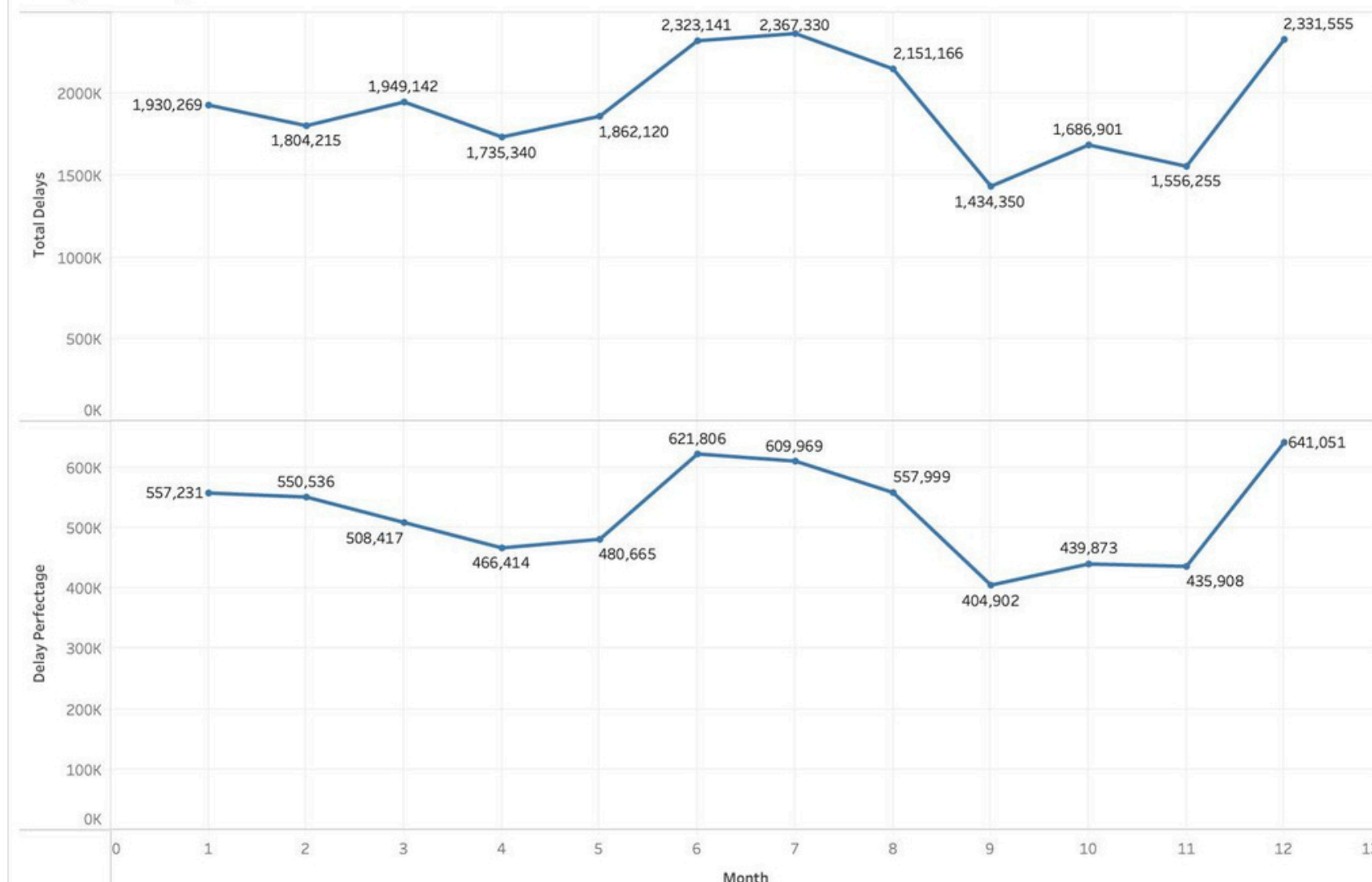


Analysis - Negative Reveiw Topics



Analysis - Delay Trend by Month

Delay Trend by Month



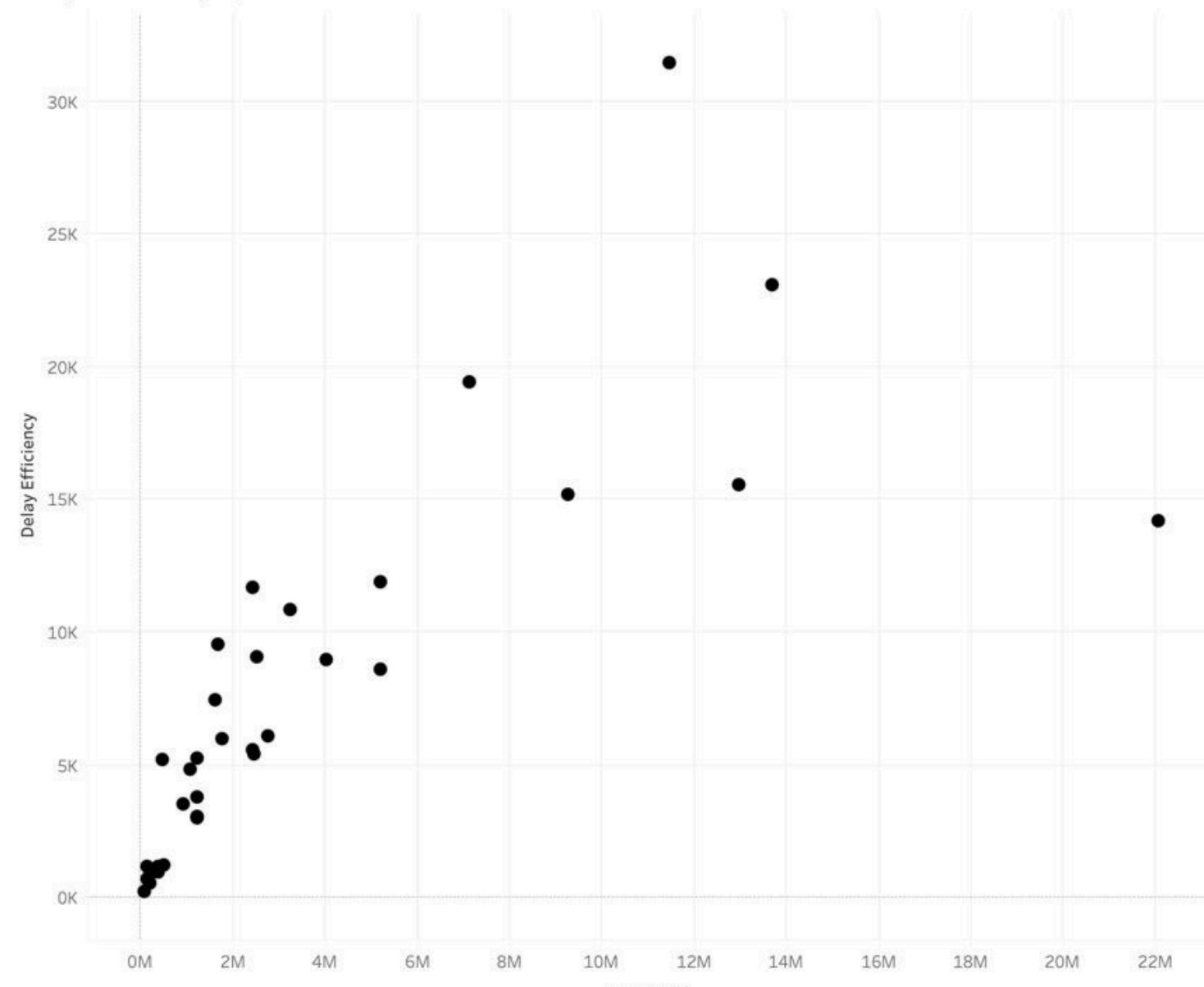
- The analysis shows significant monthly fluctuations in total delays:
 - Peak delays observed in summer months (July and August).
 - Reduction in delays during late fall and winter.
- Total delays and delay percentages often correlate with travel seasons and operational challenges.

Insights:

- Airlines could prepare additional resources during peak delay months.
- Seasonal weather conditions are a major contributor to delay spikes.

Analysis - Delay Efficiency by Airlines

Delay efficiency by Airlines



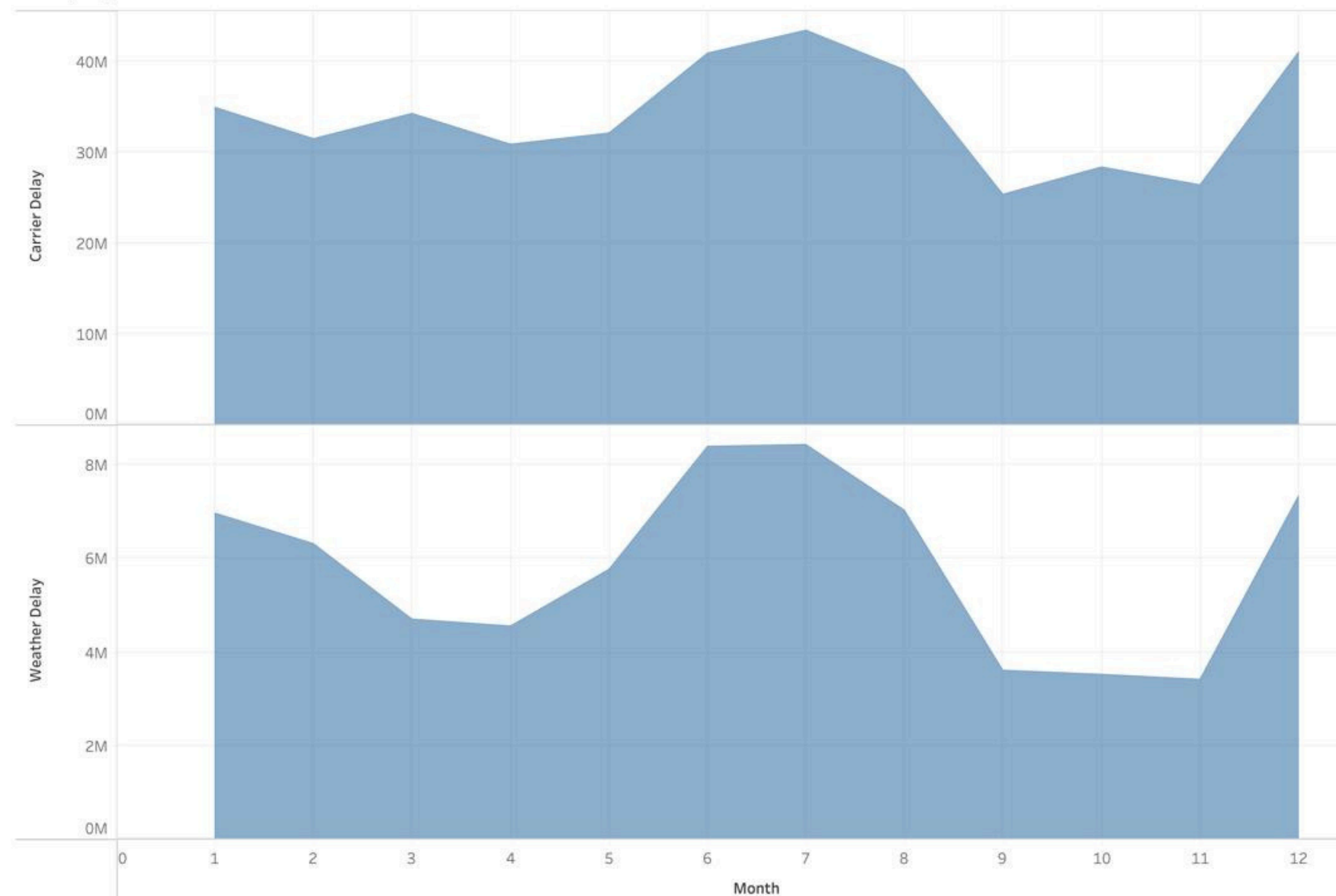
- Airlines with higher total arrivals demonstrate varied delay efficiencies.
- Some airlines maintain lower delays despite high volumes, indicating operational efficiency.

Insights:

- Airlines with higher delay efficiency should act as benchmarks for others.
- Highlight inefficiencies to assist in operational improvements.

Analysis - Delay by Cause

Delay by Month



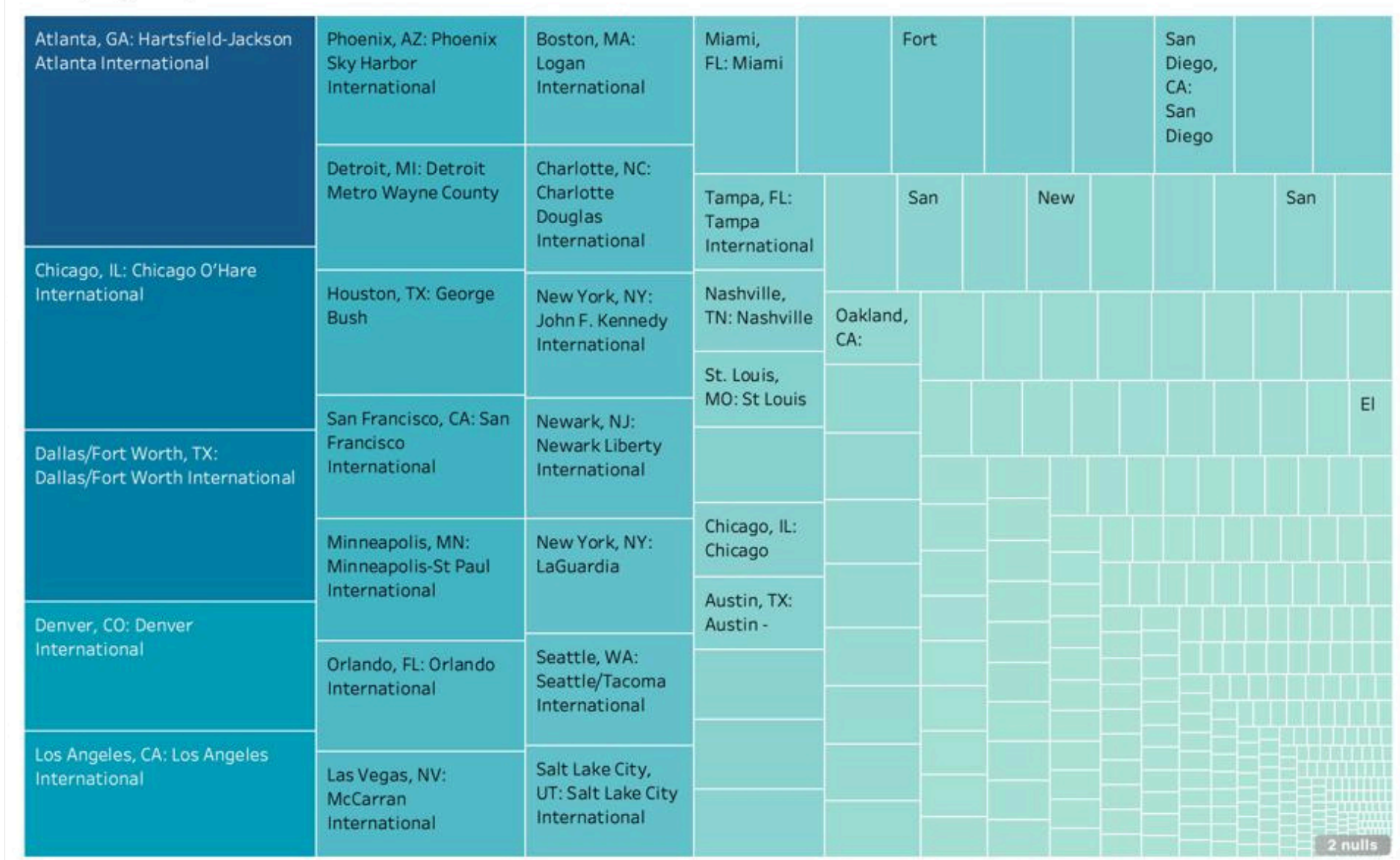
- Carrier-related delays significantly overshadow weather-related delays in most months.
- Weather delays spike in specific months (e.g., winter storms or summer thunderstorms).

Insights:

- Carrier-related delays require internal operational improvements.
- Weather delays emphasize the need for contingency planning.

Analysis - Delay by Airport

Delay by Airport



- Airports like Hartsfield-Jackson Atlanta and Chicago O'Hare show the highest delays.
- Smaller airports tend to have significantly fewer delays.

Insights:

- Large hub airports experience greater operational bottlenecks.
- Investments in airport-specific infrastructure can mitigate delays.

Analysis - Airline-Specific Trends

Delay by Airlines



- Airlines like JetBlue, Southwest, and United have higher delay volumes, reflecting operational challenges.
- Airlines with lower delays could serve as models for efficient scheduling and management.

Insights:

- Operational strategies for delay reduction should prioritize frequent carriers with high delays.
- Targeted optimization programs for these airlines can improve overall customer experience.

Recommendations

- Focus on high-delay months for strategic improvements.
- Address carrier-specific inefficiencies and adopt best practices from efficient airlines.
- Invest in predictive tools and infrastructure upgrades to minimize delays.

Proposed Solutions

Switch to Dark Mode

Customer Reviews

Search reviews...

Sort By

Anonymous

Delayed flight out of Miami over two hours causing us to miss connecting flight to Portland. Rebooked automatically to have us fly to Phoenix and be stranded overnight and take a flight to Portland Wednesday 11/13 in the morning. Flight in Miami delayed again so we would miss our rebooked connecting flight in Miami to phoenix so stranded in Miami overnight. Rebooked flight from Miami to PDX but it is not a nonstop like original, have a layover in Texas. American Airlines Booked us a hotel overnight only to show up and have them tell us there is no rooms available, this forced us to go back to airport to stay there overnight with our two kids. This forced us to pay for an Uber back to the airport. In total we are now going to arrive home more than 16 hours later from the original time that was booked. Loss of wages for two people, pet sitter additional day and night, uber rides to hotel, both kids miss school, extra day of parking fees at airport

★

Unflag

Anonymous

This complaint is due to a mechanical problem which caused me to miss my connecting flight. I stood in line for 45 minutes at the help desk to receive a voucher as promised. Upon receipt of a free hotel and a \$12.00 food credit, I left to pick up my luggage, which was pulled at my originating airport to be picked up in Dallas ft worth since I knew ahead of time I would have to spend the night in Dallas and was told I would receive a voucher for my hotel along with my luggage. I waited for all the bags to come off and my luggage was not there as promised. I went out to call the hotel, and the phone just went voicemail after numerous attempts. I googled the address of the hotel, and could not get anyone to answer to even get a ride to the hotel. After having this happen to me on a previous trip with American Airlines, where I received a voucher to a hotel, got a bus to the hotel and there were no reservations available. I realized I did not want To pay for an uber to a hotel that clearly either wasn't in business anymore or were overbooked. I

Anonymous

I had an incredibly disappointing experience with American Airlines On November 5th landed from PVD to PHL. We were told to standby for the connecting flight to MBJ. After spending over 5 hours in the airport gate lounge. They canceled the flight and rescheduled a flight for November 6th. The airline staff was very rude and did not offer any suggestions or a hotel accommodations. Suggested we collect our luggage and re-board/book the next morning. The group reported to baggage claim. The attendant strongly suggested do NOT checkout your luggage. The luggage will follow on to the next flight. As well as, suggested the Airport Marriott. Appreciated his input. American Airlines already made up their minds to rescheduled a flight for November 6th. Disregarding customer service or respect.

★

Flag

Real-Time Review Analysis System:

- Flags urgent customer concerns, such as complaints about delays or poor service.

Flagged Reviews

Anonymous: Delayed flight out of Miami over two hours causing us to miss connecting flight to Portland. Rebooked automatically to have us fly to Phoenix and be stranded overnight and take a flight to Portland Wednesday 11/13 in the morning. Flight in Miami delayed again so we would miss our rebooked connecting flight in Miami to phoenix so stranded in Miami overnight. Rebooked flight from Miami to PDX but it is not a nonstop like original, have a layover in Texas. American Airlines Booked us a hotel overnight only to show up and have them tell us there is no rooms available, this forced us to go back to airport to stay there overnight with our two kids. This forced us to pay for an Uber back to the airport. In total we are now going to arrive home more than 16 hours later from the original time that was booked. Loss of wages for two people, pet sitter additional day and night, uber rides to hotel, both kids miss school, extra day of parking fees at airport

Anonymous: We started our trip home with American Airlines with a delay, we got there early, well you cant check your bags until 4 hours before flights. We're early because of having to drop a car off and wasn't sure how long that would take. So we ended up having to sit on the cold floor, because there was limited seating . After a long delay we finally reach Chicago airport, get to our gate and we find out we're delayed again, according to the pilot they sent our crew some where else, I'm sure someone knew way before we were informed, leaving us wondering who's flight was more important then ours! They took our crew, but no one seemed to think hey get another crew right away. Finally a crew shows up and as I'm checking in there was a woman calling for help because an older man almost collapsed asking for a wheel chair to help him board the plane , all 4 staff just stood there and didn't even rush to help a customer! The guy taking my ticket told her His words not mind "Jesus Christ just give us a break" that's when I walked on the plane. I contacted customer service and even though this was the Airlines fault the best they could was a \$ 50 trip voucher. How insulting is that! Worst part of our trip was flying with American airline!

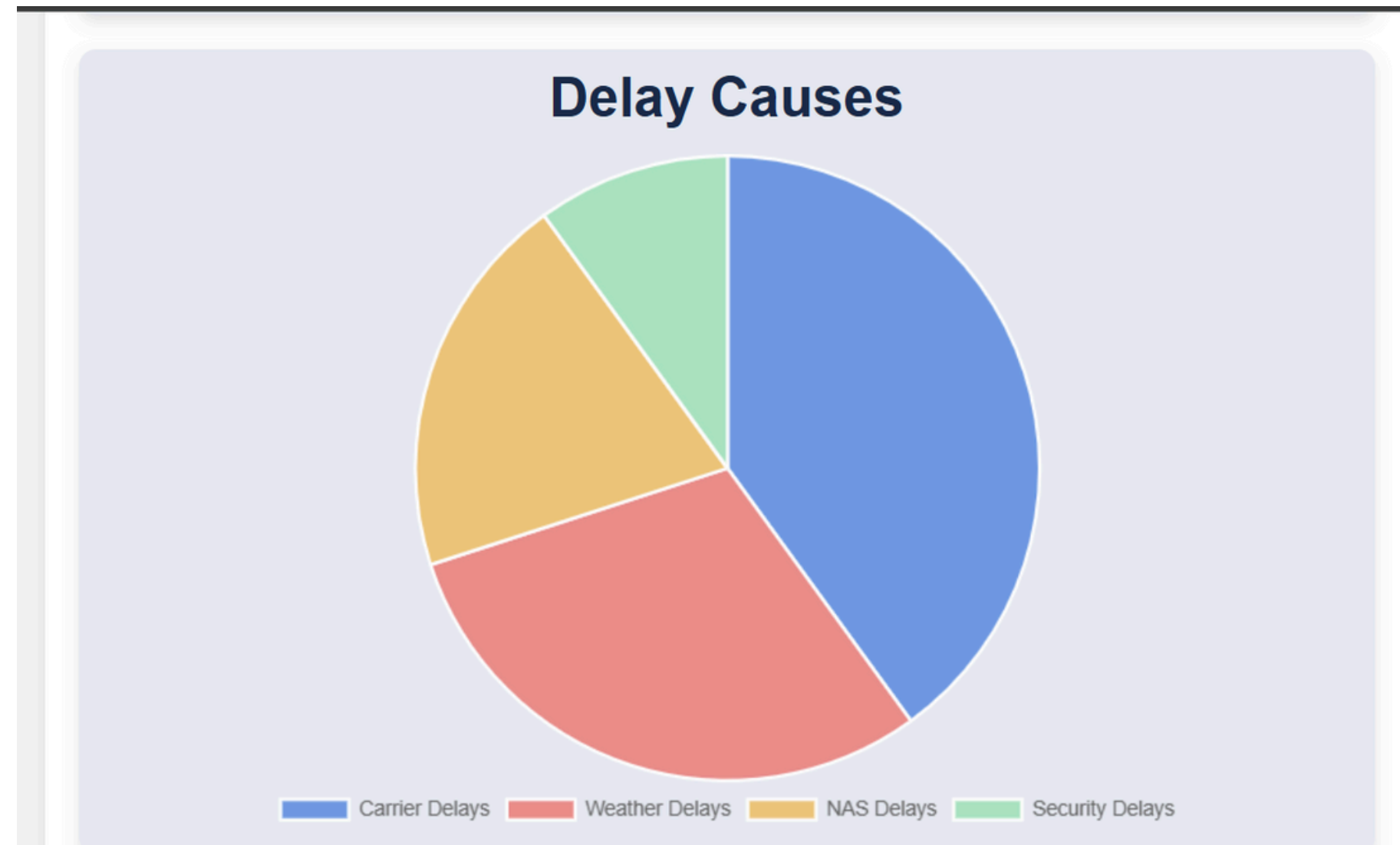
Anonymous: Not much I can say that hasn't already been said, other than that it's all true. I've been a frequent flyer for 23 years now, and I can confidently state that of all the airlines I've flown, AA by far has the worst customer service, even within the US, which is already a very low bar. Delays, cancellations, and other problems are bound to happen with any airline, but the flippant, glib and couldn't care less attitude with which these are handled by the AA staff is infuriating. Stay away.

Anonymous: American 'delayed' Airlines does it again! An hour and a half delay, with annoyed employees and an email I received that said the gate changed when it didn't. Why do I continue to fly with this miserable airline? Well, if there is a specific airport I want to fly out of, on a specific day and time, sometimes AA is the only choice I have.

•Sends alerts to airline staff for immediate intervention.

Submit Your Review

Proposed Solutions

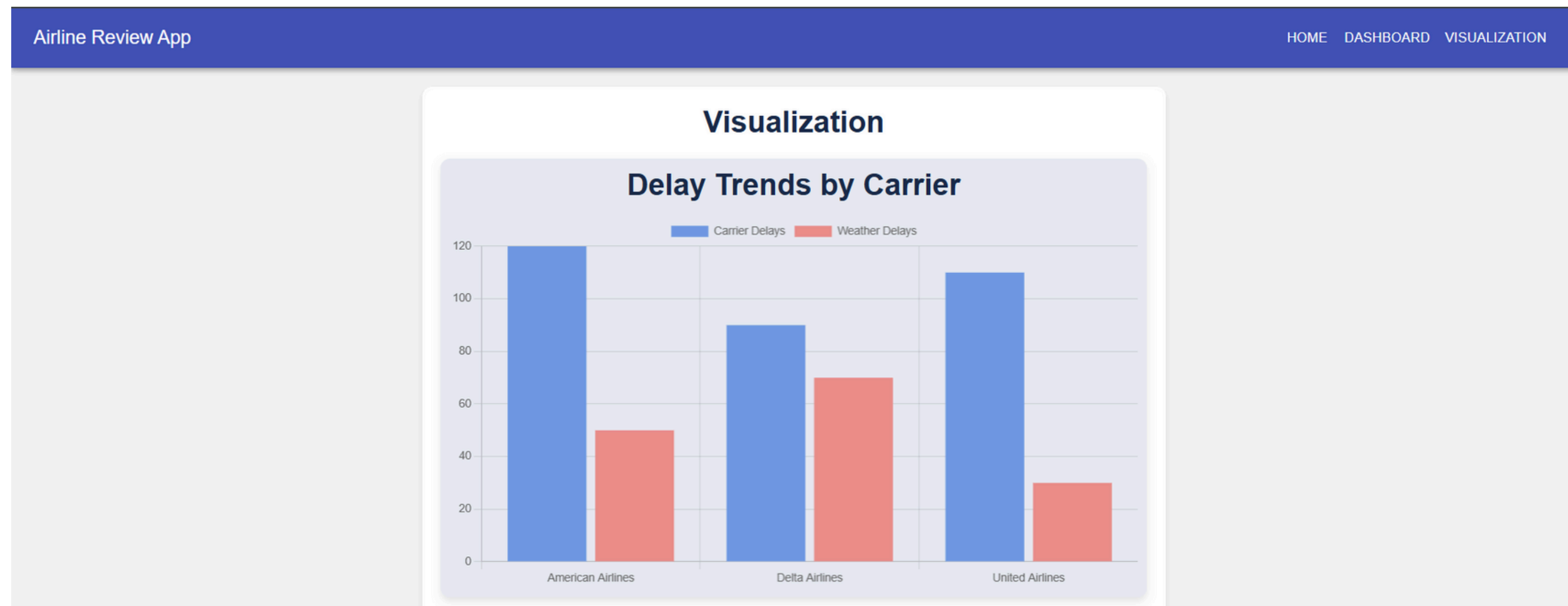


Integration with Operational Data:

- Incorporates factors like flight delays and weather conditions.

Proposed Solutions

- Provides airlines with actionable insights to improve services and prevent escalation.



Conclusion

- This project bridges the gap between customer feedback and operational insights, empowering airlines to proactively enhance their services.
- By addressing customer concerns in real time, airlines can improve satisfaction, prevent negative feedback, and strengthen brand loyalty.
- Delays are influenced by seasonal factors, airport capacity, and carrier operations.
- Carrier-related delays dominate, necessitating operational improvements.
- Weather remains a significant challenge, requiring predictive measures and contingency planning.

**Thank
You**

Questions

